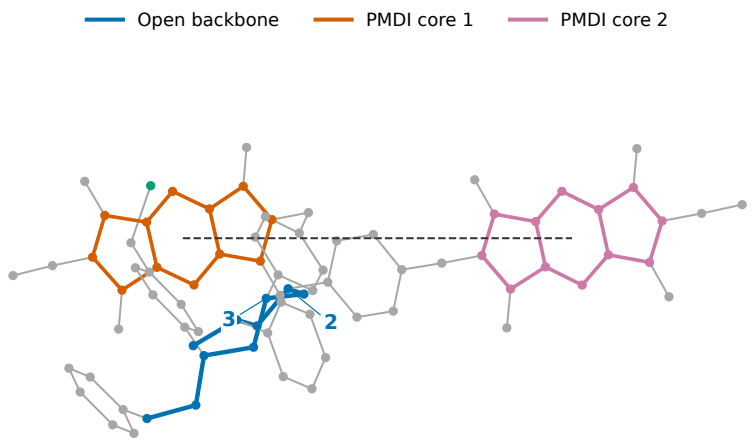
From redox molecules to grafted polymers

Which molecular advantages survive attachment and nearby redox groups?



3D core-centroid distance: 16.30 Å; plane angle: 69.0°

Actual computed PMDI-grafted open-chain model

Select a useful material, not just a promising molecule.

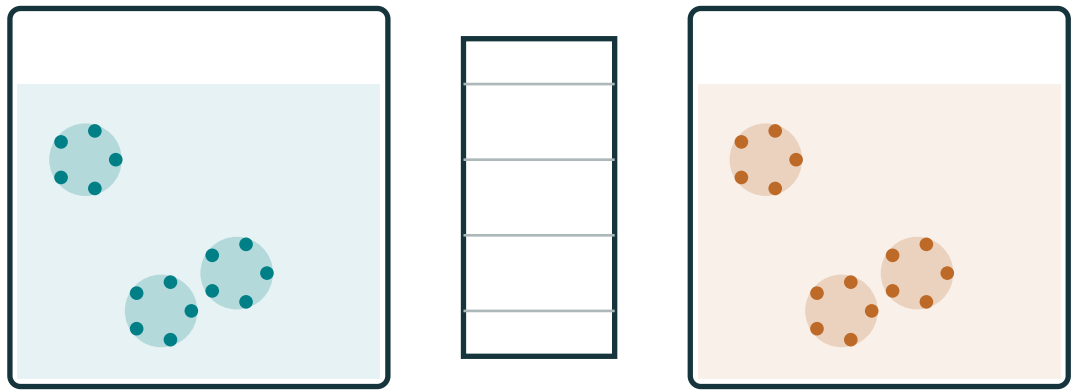
Larger redox carriers can aid separation

Store charge on redox motifs.

Use a larger carrier to aid separation.

Route: attach motifs
to a preformed polymer
with benzyl chloride sites.

Conceptual flow-battery architecture



Redox carriers | size-selective separator

The design spans redox chemistry and polymer context.

Context: Janoschka et al., Nature 2015, doi:10.1038/nature15746; Montoto et al., JACS 2016, doi:10.1021/jacs.6b06365. Cartoon is illustrative.

A motif must remain useful after grafting

01

Useful redox properties

Voltage window
Accessible capacity
Stable charge states

02

Feasible attachment

Exact precursor
Defined reaction conditions
Core retained after grafting

03

Weak motif interactions

Spacing and conformation
Motif contact strength
Intersite coupling

A candidate must pass every required axis.

Route screening links UMA starts to solution DFT

Exact models

Exact graphs
Precursor/product
Oligomer occupancy

→

Route review

Route evidence
Defined conditions
Screen before QM

→

UMA

Gas-phase ranking
Seeds to relaxed poses
Charge and spin

→

DFT (MeCN)

Optimize geometry
Electronic energies
Hessian + qRRHO

Molecular response

E_1, E_2 vs Fc/Fc^+
Directional solution response

Grafting ΔG

ΔG_{graft}
Defined base and byproducts

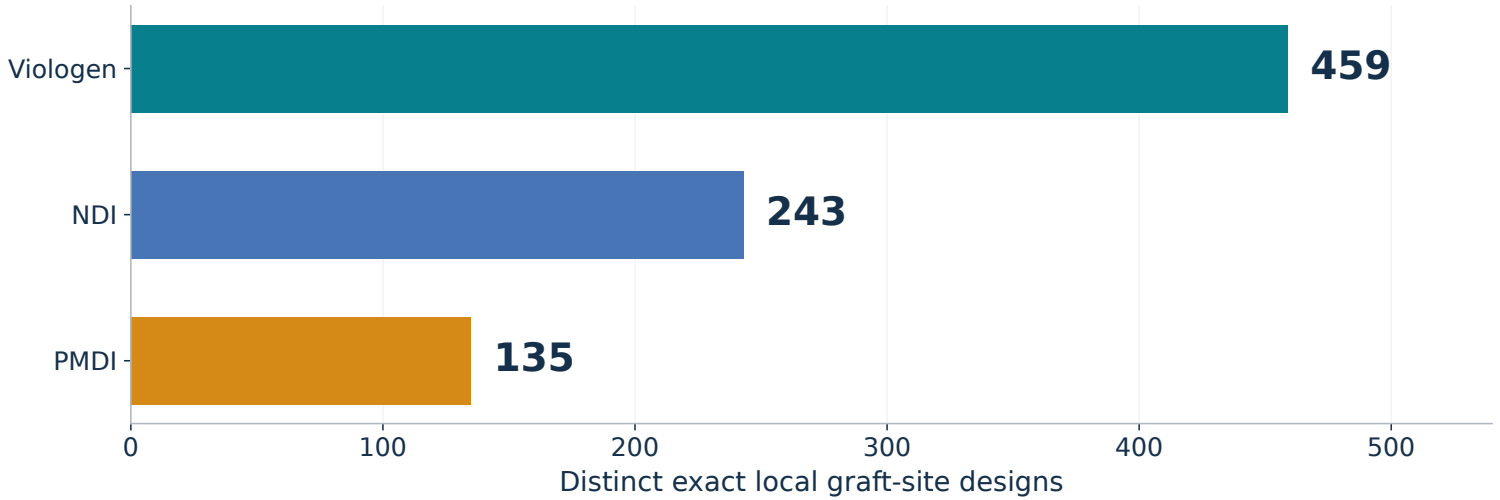
Neighbor effects

Neighbor $\Delta\Delta G$
Local interaction E_{int}

UMA speeds the search; checked DFT supports the decision.

UMA is a gas-phase structural accelerator, not the source of the reported solution free energies. Each exact polymer graph is evaluated separately. The tested Nitro force-repeatability gate fails; the standard-inference/V100 profile remains separate.

A systematic grid anchors a 912-graph pool



912

exact graphs in the registry

Design variables

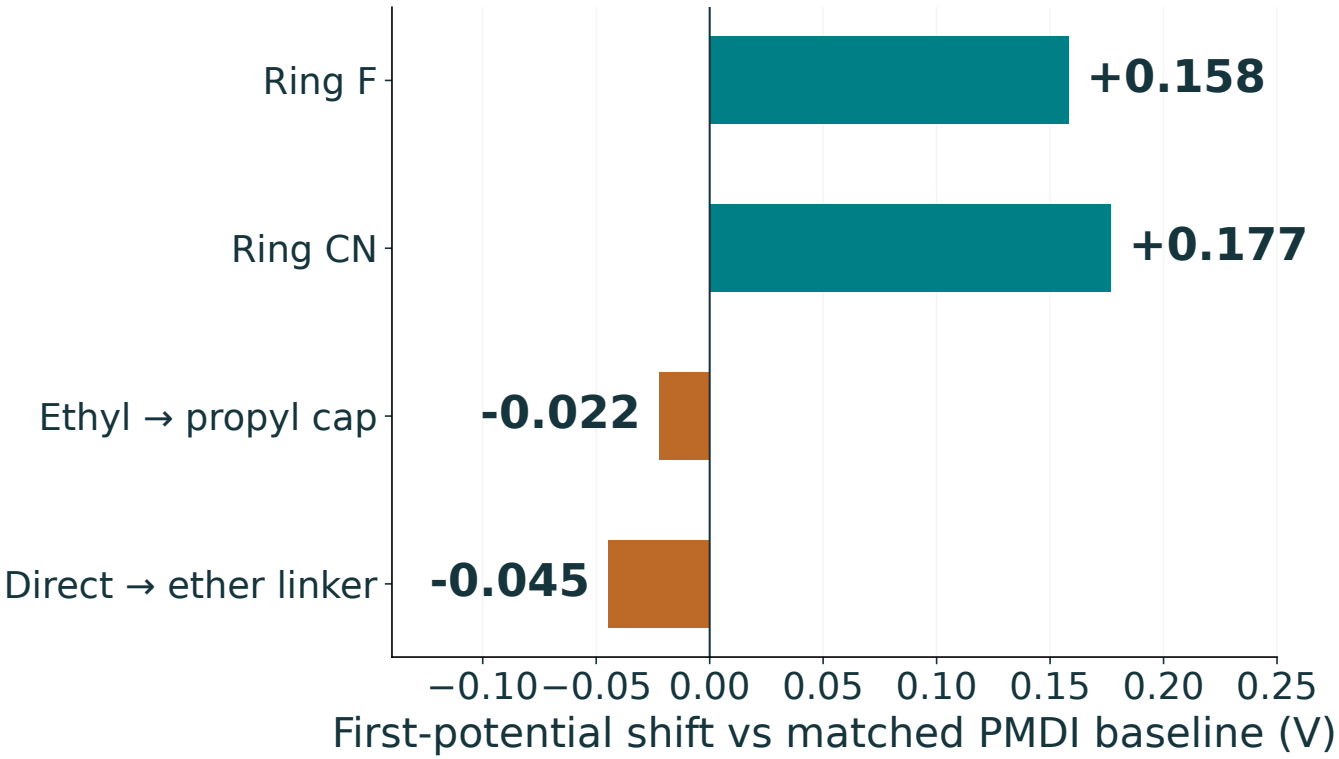
Core
Linker
Cap
Ring substituent

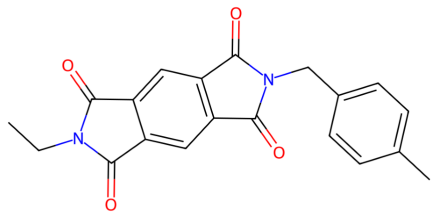
The 837-grid subset spans core, linker, cap and ring choices.

Enumeration feeds exact-route and quantum checks.

The systematic subset contains 459 viologens, 135 PMDIs and 243 NDIs. The unified registry has 912 graphs and 3 reviewed baseline routes. Eight new direct N-linked phenothiazine F/CN designs remain held and uncomputed. Graph/route QC is separate from quantum or material validation.

PMDI voltage responds to ring substitution





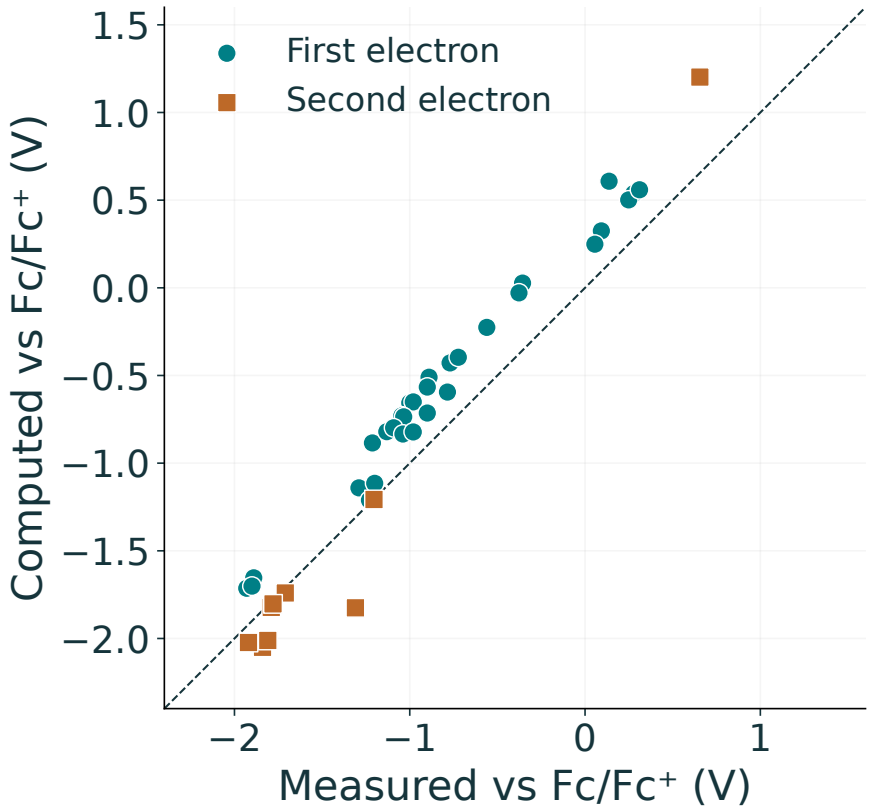
Same baseline

PMDI core
Ethyl cap
Local benzyl graft

Matched comparisons identify molecular tuning levers.

Computed first reductions in MeCN, relative to the exact baseline. Four illustrative contrasts from 18 systematic three-core comparisons; all 26 cohort contrasts reconstruct.

The benchmark error scale is about 0.25 V



0.247 V

Unfitted mean absolute error

39 endpoints

30 molecules

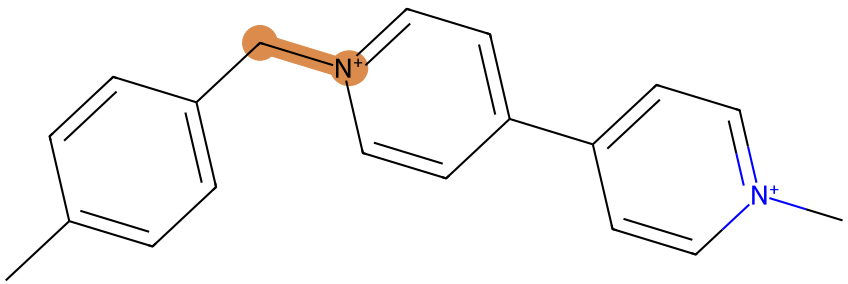
7 studies

MeCN; RMSE = 0.279 V

The benchmark defines a limited molecular accuracy context.

Retrospective exact-free-molecule panel. Molecule-cluster 95% MAE interval: 0.213–0.286 V. No NDI endpoint; grafted-polymer accuracy is not established.

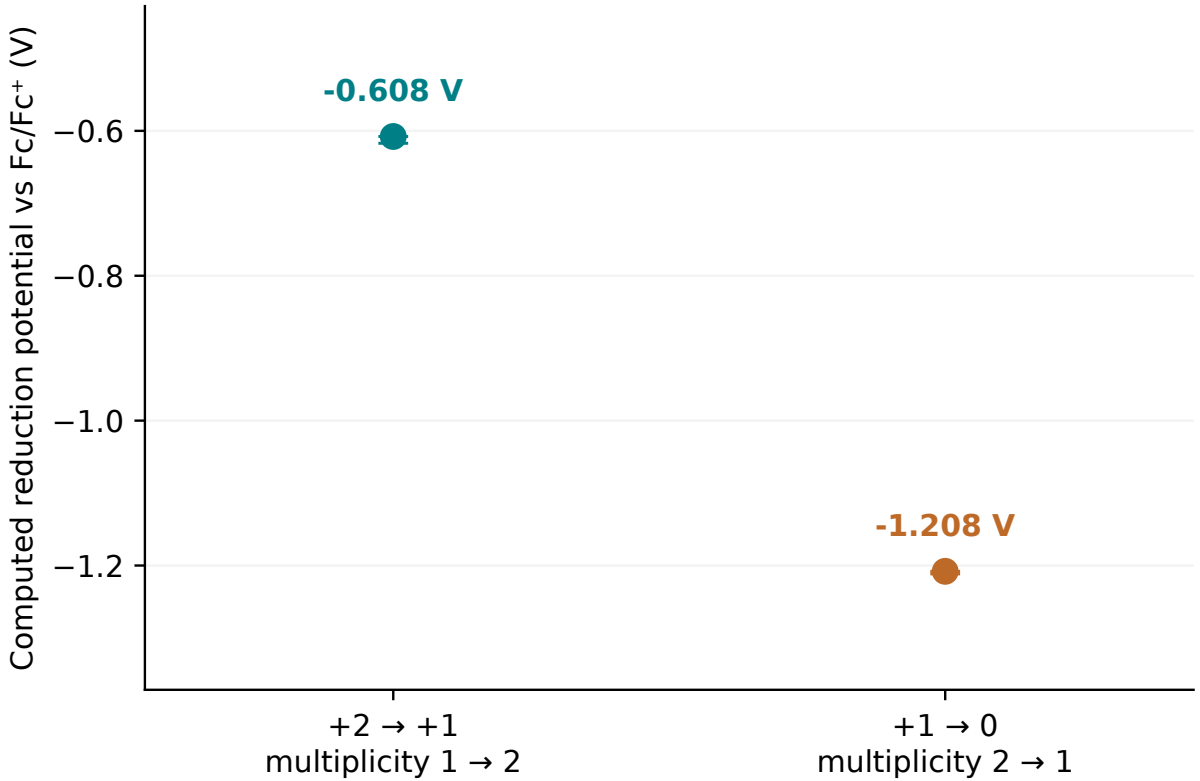
One exact viologen links redox and grafting



Exact capped graft graph

Grafting $\Delta G = -11.49\,\mathrm{kJ/mol}$

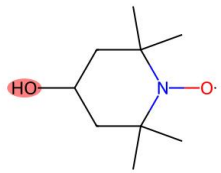
Monomer $\Delta G_{\mathrm{disp}} = +57.91\,\mathrm{kJ/mol}$



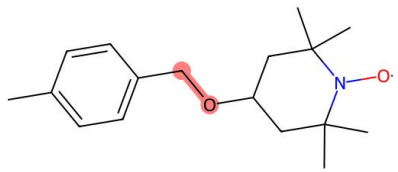
One graph connects enumeration, route, redox and grafting.

Capped local model, MeCN, 298.15 K; three native states and matched Fc, no fitted offset. Bars show rotor-setting sensitivity. Grafting is a finite local-site separated-ion model; chloride contacts are tested separately; occupied ensembles and coupling remain pending. Drawing shows resting connectivity; no charge/spin localization is inferred.

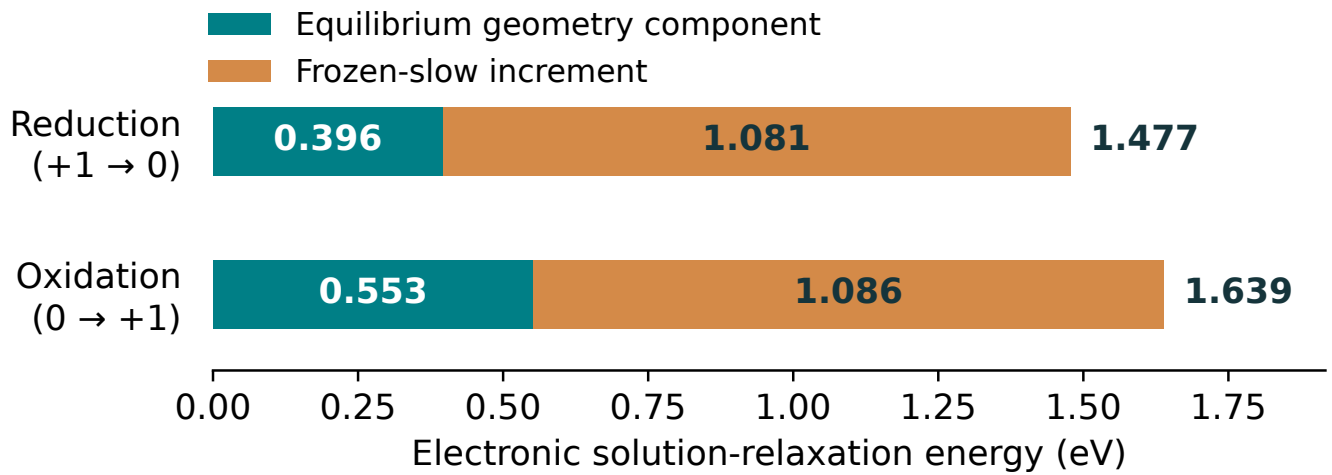
TEMPO needs a directional response model



4-Hydroxy-TEMPO precursor



4-Methylbenzyl local graft model



Reviewed attachment route

$E_{OX} = +0.579$ V vs Fc

Directional asymmetry: **10.4%**

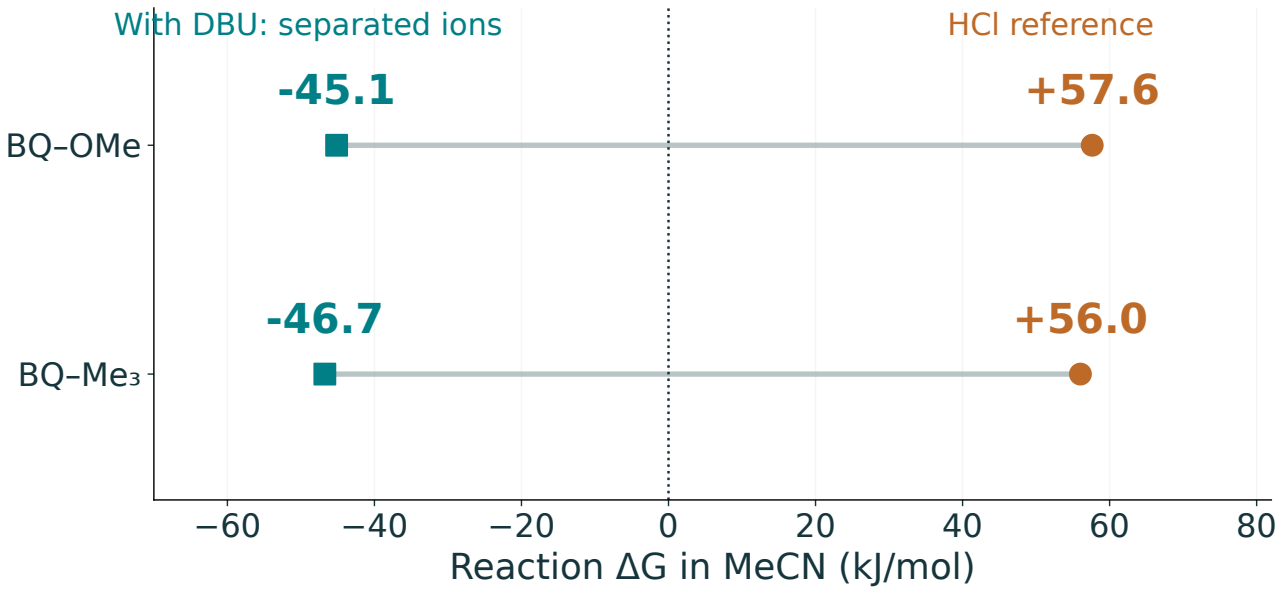
Above the declared 5% diagnostic

Keep both directions; one symmetric parameter is unqualified.

Exact capped TEMPO graft; native-SMD frozen-slow electronic response in MeCN at two checked geometry-level minima. Not grafting free energy or a measured rate. No gas-inner or Born-outer term is added.

The base defines the grafting reaction

$$\text{Motif–OH} + \text{polymer–CH}_2\text{Cl} + \text{B} \longrightarrow \text{polymer–CH}_2\text{–O–motif} + \text{BH}^+ + \text{Cl}^-$$

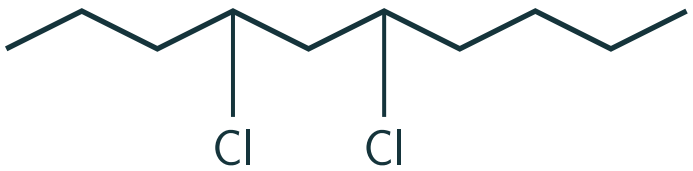


Base-required BQs remain candidates for further evaluation.

Local graft-site calculations, MeCN, 298.15 K. Computed DBU capture is a shared offset of −102.78 kJ/mol; rankings are unchanged. pKa validation exposes an approximately 23 kJ/mol discrepancy. Signs remain conditional; selectivity and yield are untested.

Four states isolate the neighbor effect

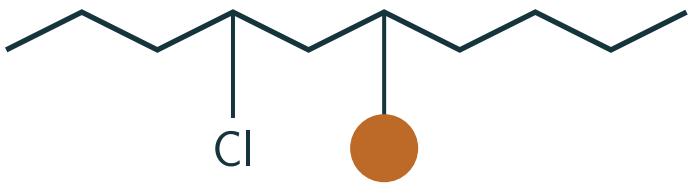
G_{00}



G_{10}



G_{01}



G_{11}

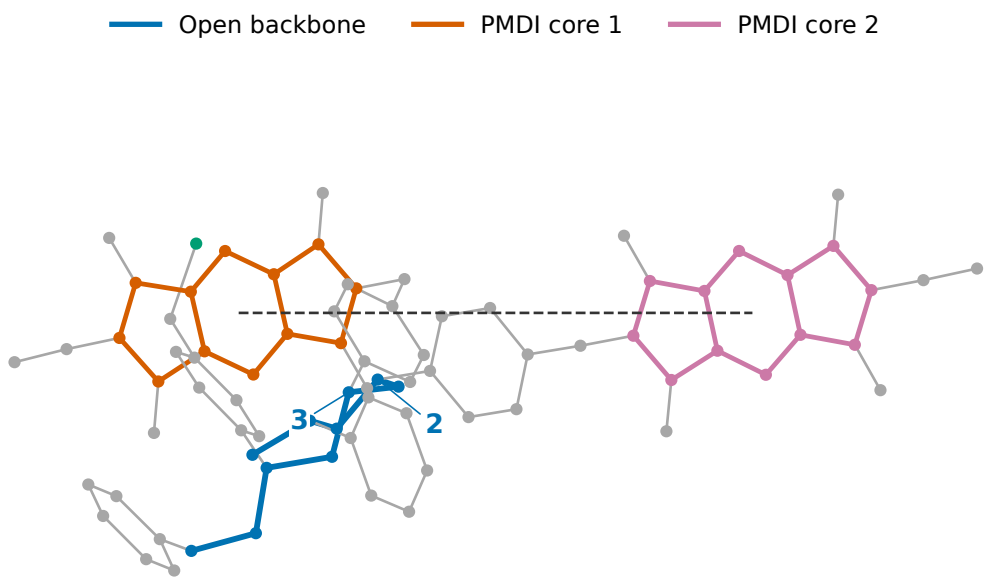


$$\Delta\Delta G = G_{11} + G_{00} - G_{10} - G_{01}$$

Measure how a nearby graft changes the next grafting step.

Schematic of the exact occupancy cycle, not an atomistic drawing. All backbones are open chains. Cl denotes an unreacted benzyl-chloride handle; colored circles denote grafts. Actual loading and spacing are unknown.

A sampled PMDI neighbor has a modest effect



3D core-centroid distance: 16.30 Å; plane angle: 69.0°

−1.86

kJ/mol
Neighbor $\Delta\Delta G$

16.30 Å

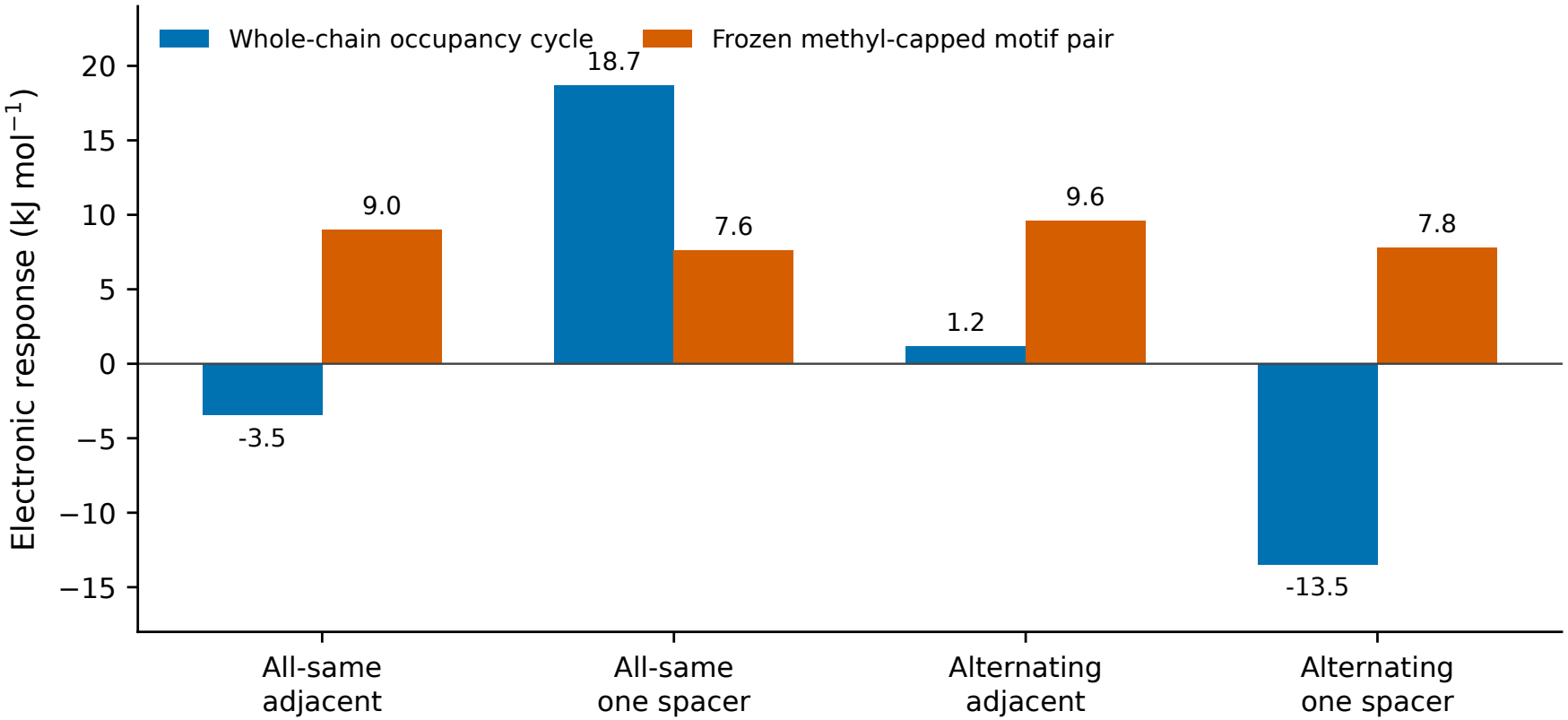
Core separation

Four matched free energies; one minimum per graph.

The finite PMDI cycle quantifies a polymer-context effect.

Exact ethyl-PMDI design. Local redox: $E_1 = -1.076$, $E_2 = -1.977$ V vs Fc. First-site attachment is 5.34/6.11 kJ/mol less favorable than the capped model. Whole-chain nonadditivity includes backbone and solvent response.

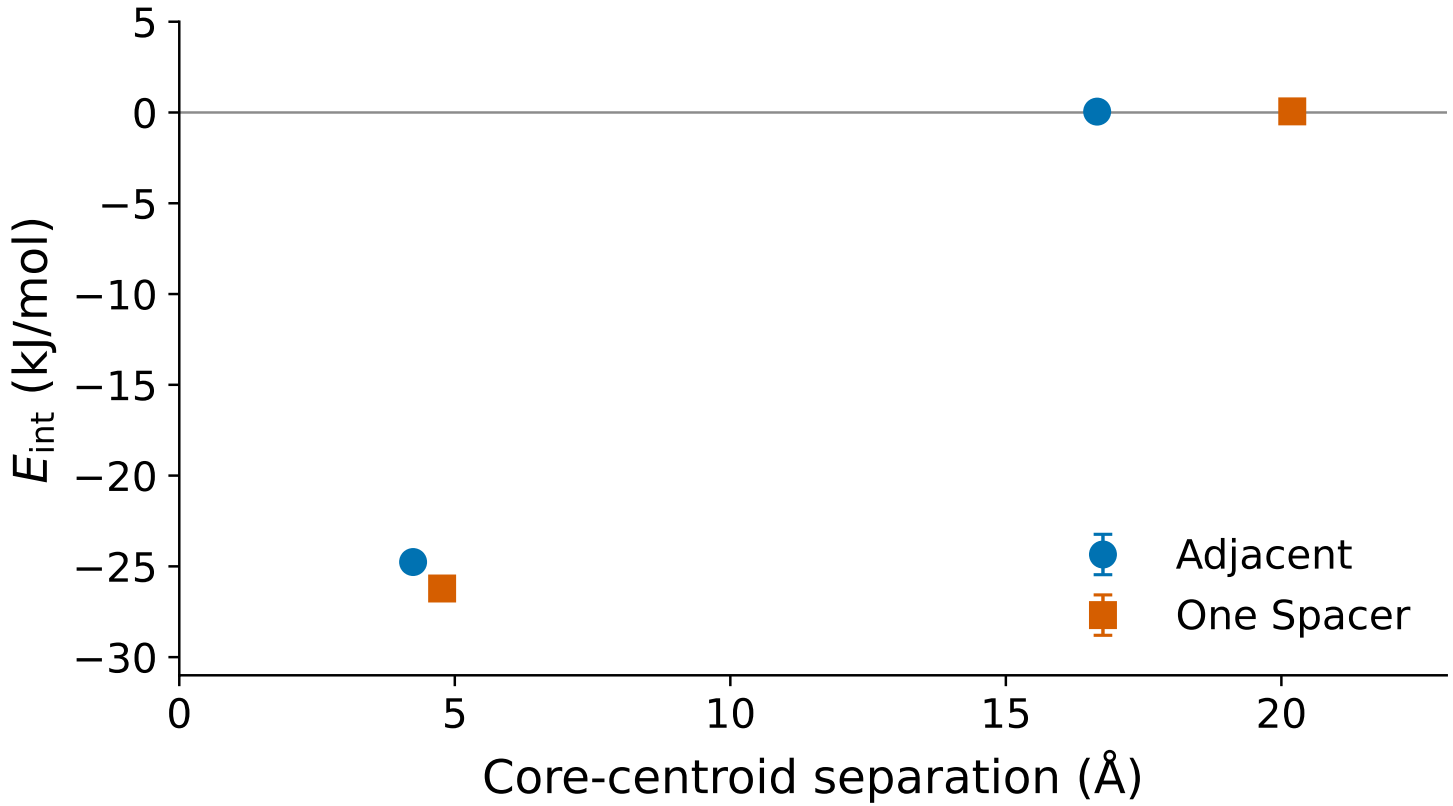
A favorable chain cycle can coexist with motif repulsion



The two electronic definitions have opposite signs in two of four frames.

Twelve A/B/AB triplets and 72 actual SCFs. Frozen methyl caps at 1.07/1.09/1.11 Angstrom; MeCN cap span below 0.003 kJ/mol. Same corrected r^{2}SCAN-D4/SVP smooth-surface model. No binding G, pure-backbone decomposition, coupling or populations.

A spacer can still permit close contact



−25 to −26

kJ/mol
Close phthalimide poses

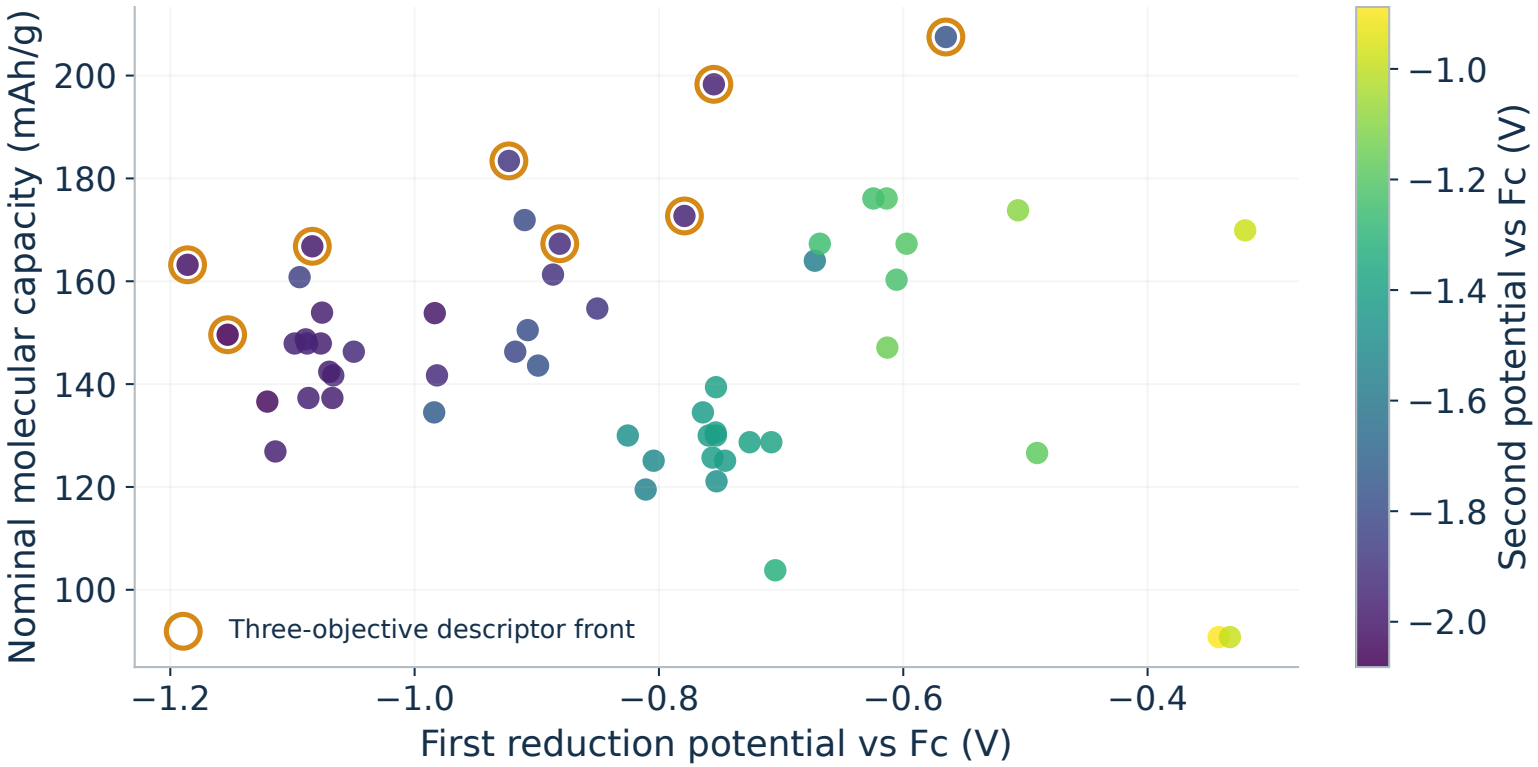
≈ 0

kJ/mol
Separated poses

A sequence spacer does not guarantee weak contact.

Frozen capped-fragment electronic interactions in MeCN, $E_{AB} - E_A - E_B$. Whiskers show cap-length sensitivity, not statistical intervals. Phthalimide is a control. No association free energy, population, basis-superposition correction or electronic coupling is established.

Every required axis must pass



8

Molecular descriptor leaders

0

Demonstrated all-axis candidates

Descriptor leaders still need full material assessment.

Projection of the three-objective comparison over 56 reduction-active local models. Nominal molecular capacity excludes polymer/loading penalties. Route, stability, accessibility, occupancy and interaction evidence remain incomplete.

Three lessons guide the next designs

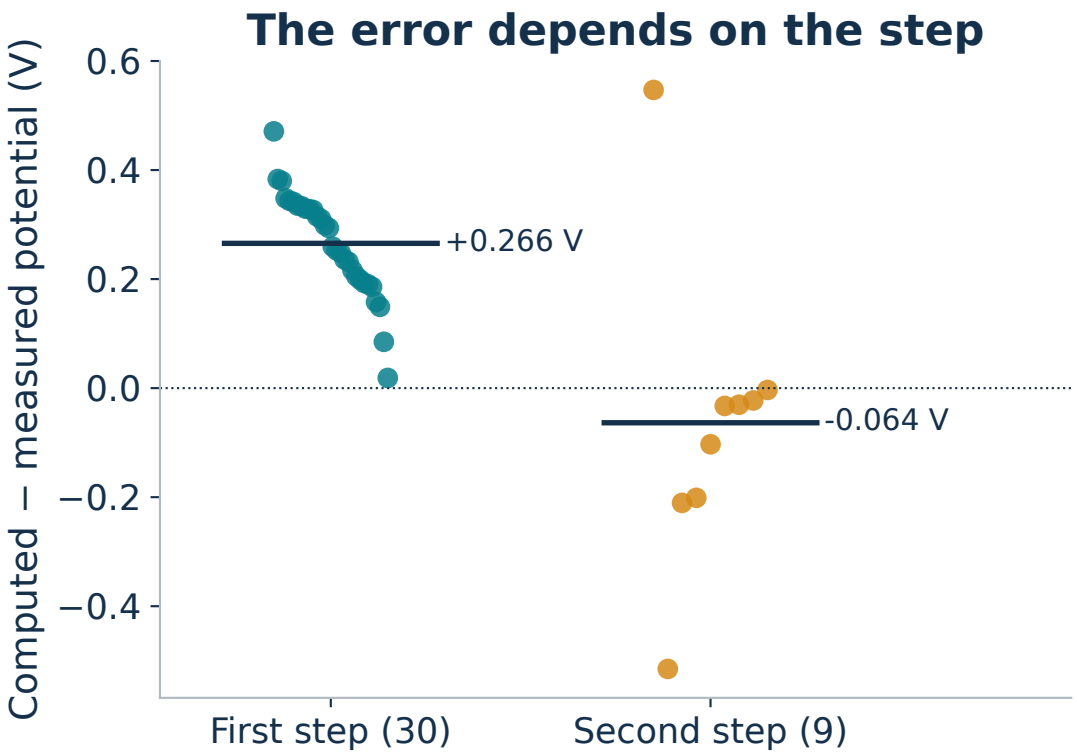
1 **Tune redox chemistry systematically.**
Ring substitutions shift voltage; benchmark error is about 0.25 V.

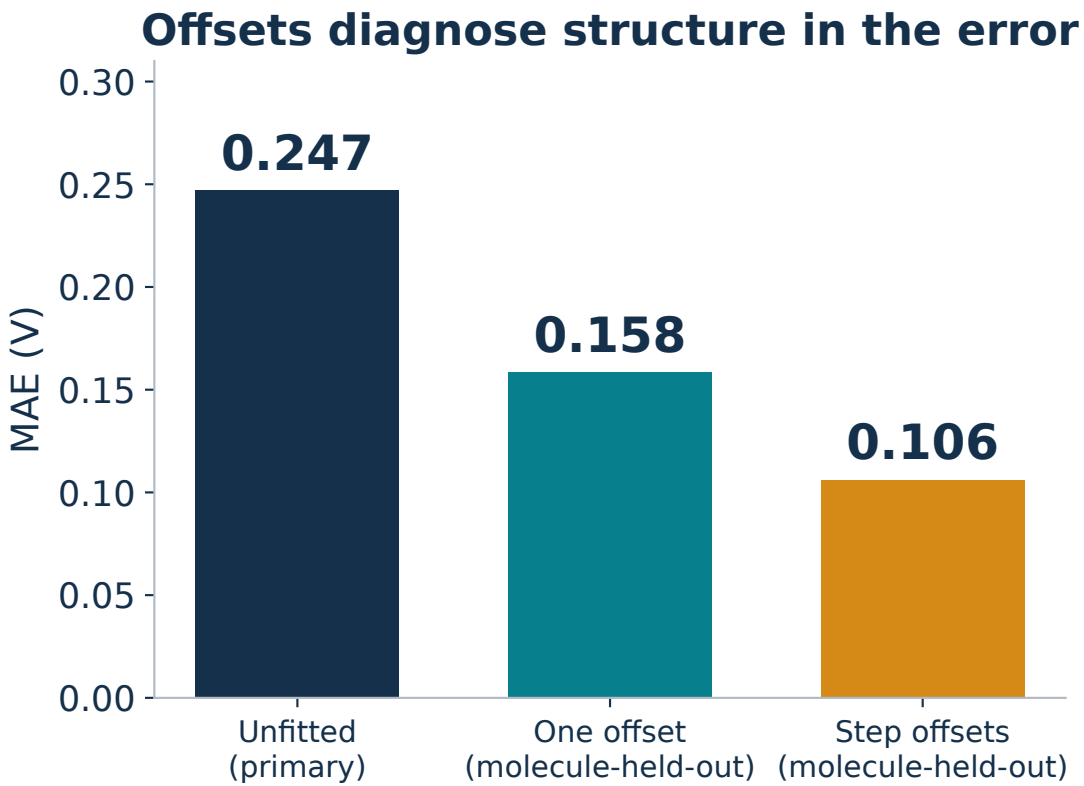
2 **Evaluate a defined attachment reaction.**
Base and speciation affect feasibility; a reference reaction is insufficient.

3 **Distinguish chain response from motif interaction.**
Spacing and conformation change the net chain response.

Next: broader chemistry and complete polymer-context evidence.

Benchmark bias and numerical sensitivity





Offset diagnostics supplement the unfitted benchmark.

Molecule-held-out MAE: 0.158 V, one offset; 0.106 V, separate step offsets. Five neutral SVPD controls move potentials by −11.7 to −35.1 mV. Neither diagnoses all remaining error sources.

Computed values and external validation differ

Quantity	Evidence	Limit
Redox potential	39 exact-parent MeCN endpoints	No polymer or NDI accuracy claim
Grafting ΔG	Balanced reactions; experimental base-pKa comparisons	Speciation, selectivity, barriers
Polymer neighbor effect	Finite PMDI and viologen cycles	Broader ensembles pending
Local motif interaction	12 triplets / 36 electronic states	No binding G or coupling

Keep numerical checks, accuracy and sampling distinct.

Both empty backbones pass the batch check

Declared expansion

20 starts for each of
12 graph/configuration sets.

Two backbone configurations.
Occupancy and spacing varied.

Both empty components

20 checked inputs each.
20 / 19 distinct minima.

10 → 20 ensemble changes:
−1.03 / −1.23 kJ/mol.

Both empty components pass; occupied cycles remain open.

The new analysis uses a separately identified smooth-surface MeCN model; it cannot be mixed with the primary PMDI cycle. Global sampling convergence is not established.

The retained chain context changes the neighbor response

Three checked finite G cycles

All-same, adjacent: −3.94 kJ/mol

All-same, one spacer: 22.58

Alternating, one spacer: −5.30

Matched atom, charge and reference balance

What is qualified

Three of four finite ionic cycles

Eleven checked local minima

Own Hessian/nuclear term per state
One cycle and ensembles pending

Finite whole-chain response is separate from motif binding or coupling.

Corrected r^2SCAN-D4/SVP, smooth-surface SMD MeCN, 298.15 K. Each state has its own Hessian/nuclear terms; no explicit spectator ions. No primary molecular-model substitution or material admission. Source: immutable ionic-cycle evidence.

Explicit chloride contacts test the ion convention

$V^{2+} + Cl^- \rightarrow VCl^+$ Total contact state: $+1$, singlet

Checked outcome	$\Delta G^\circ_{\mathrm{assoc}}$ (kJ/mol)
N0 above, minus mode	15.61
N0 above, plus mode	15.42
N1 above	22.36
N1 below	22.11

Four outcomes

Three duplicate groups

1 M, MeCN, 298.15 K

Other contacts pending
No population weights

Finite chloride association is separate from covalent grafting.

Primary composite MeCN, 298.15 K; each contact has its own native Hessian. N0: benzyl-bearing N; N1: methyl-bearing N. One-atmosphere to 1-M correction: −7.93 kJ/mol, once. N0 outcomes share one group (heavy RMSD 0.00377 Angstrom; energy tolerance 0.2 kJ/mol). Other ions, populations and yield remain unresolved.

The exact molecular free-energy cohort reproduces

New numerical evidence

Nine molecular Hessians

81 SCF frames and orbital-spin checks

One analytic chloride partition
Matched electronic and nuclear terms

Six complete cycles

Potential changes $< 2.45 \times 10^{-8}$ V

Reaction changes $< 1.55 \times 10^{-6}$ kJ/mol

Same source geometries and protocol
Three rotor settings retained

Numerical reproduction and experimental accuracy remain separate.

Corrected primary-model exact viologen cohort: two Fc-referenced reductions, three precursor-specific quaternizations and disproportionation. r^{2}SCAN-D4/SVP Hessians plus ωB97M-V/TZVPD electronic terms in SMD MeCN. No polymer-cycle or ensemble admission.

Reproducibility and model qualification remain explicit

Available

Code and exact chemical identities

Checkpointed calculations

Frozen evidence and raw-state packages
Relocated arithmetic/Hessian replay

Remaining

Public data DOI and metadata

Broader deployment portability

Nitro force-repeatability qualification
Complete material-level assessment

Solution free energies retain the checked DFT protocol.

Current H100 Nitro repeatability controls fail the stated force threshold; installation is not numerical qualification. V100 uses a separately tested standard-inference fallback.